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Textual emotion classification using MPNet and cascading broad learning

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ABSTRACT

As one of the most important tasks of natural language processing, textual emotion classification (TEC) aims to recognize and detect all emotions contained in texts. However, most existing methods are implemented using deep learning approaches, which may suffer from long training time and low convergence. Motivated by these challenges, in this paper, we provide a new solution for TEC by using cascading broad learning (CBL) and sentence embedding using a masked and permuted pre-trained language model (MPNet), named CBLMP. Texts are input into MPNet to generate sentence embedding containing emotional semantic information. CBL is adopted to improve the ability of feature extraction in texts and to enhance model performance for general broad learning, by cascading feature nodes and cascading enhancement nodes, respectively. The L-curve model is adopted to ensure the balance between under-regularization and over-regularization for regularization parameter optimization. Extensive experiments have been carried out on datasets of SMP2020-EWECT and SemEval-2019 Task 3, and the results show that CBLMP outperforms the baseline methods in TEC.

1. Introduction

As various social network platforms have become an essential medium for people (Peng et al., 2019, 2018; Wu et al., 2021), who often utilize smartphones, laptops and other equipments to release their information on them and further to express their opinions anytime and/or anywhere (Zhao et al., 2018). Thus, a huge number of texts have emerged on the social networks and most of which contain emotion. Textual emotion classification (TEC) aims to recognize and detect emotions (e.g., joy, love, anger, sadness, fear, evil, and desire) automatically from texts (Zeng et al., 2023). TEC usually includes the following three types: the methods based on the emotion lexicon (EL) (Song et al., 2015; Staiano & Guerini, 2014; Xu, Xu, Zheng, Q. Lu, Wong, & Wang, 2013), traditional machine learning (ML) (Purver & Battersby, 2012; Suttles & Ide, 2013; Zhao, Dong, Wu, & Xu, 2012), and deep learning (DL) (Abdul-Mageed & Ungar, 2017; Kim, 2014; Tafreshi & Diab, 2018; Zhang et al., 2018).

The EL-based method mainly uses EL resources to extract the keywords of emotion expression and then classifies emotion from texts. For example, Xu et al. (2013) constructed a Chinese emotion lexicon,

called WordNet-Affect, based on the English lexicon WordNet (Miller, 1995). Staiano and Guerini (2014) built an emotion lexicon, called DepecheMood, with a crowd-sourced affective annotation manner. Song et al. (2015) built a lexicon for emotion classification, called Lex-c, by combining seed words and emotions. It mainly depends on the quality and coverage of EL. Because of the continuous emergence of new online words (e.g., stan, clapback), it is difficult to construct, maintain, and expand a high-quality EL.

The ML-based method often uses supervised and semi-supervised learning algorithms, which depend on a large number of labeled data. For example, Zhao et al. (2012) designed a classifier for sentiment analysis by using naive Bayes (NB). Purver and Battersby (2012) proposed a semi-supervised emotion classification method by using support vector machines (SVMs). Suttles and Ide (2013) designed a classifier for emotion classification by using NB. However, these supervised learning algorithms require a large number of high-quality data, and need more time and manpower to annotate the corpus, which may affect its practicality (Ran, Zheng, Hailun, Weiping, & Dan, 2018).

In recent years, many methods (Cao et al., 2020) for TEC are DL-based, which usually adopt deep neural networks to automatically

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extract features in texts. The DL-based methods can effectively conquer the drawbacks of the EL-based and ML-based methods in a great extent. For example, Kim (2014) and Zhang et al. (2018) utilized convolutional neural networks (CNN) (Lecun, Bottou, Bengio, & Haffner, 1998); Abdul-Mageed and Ungar (2017) and Tafreshi and Diab (2018) exploited gated recurrent neural networks (GRNN) (Cho et al., 2014); Basile, Franco-Salvador, Pawar, Stajner, Rios, and Benajiba (2019) and Xiao (2019) employed bi-directional long short term memory (Bi-LSTM) (Schuster & Paliwal, 1997) and attention (Itti & Koch, 2001). However, its convergence speed is pretty slow and training time is long because of too many parameters and the complex structure.

Broad learning (BL) (Chen & Liu, 2018) is a flat network on the basis of random vector function-link network (RVFLN) (Pao, Park, & Sobajic, 1994). BL has some advantages including simple structure, incremental modeling, strong generalization ability, and convenient theoretical analysis. However, it is difficult to capture textual context information because of the feed-forward network structure of BL. In addition, although the pre-trained language model like BERT (Devlin, Chang, Lee, & Toutanova, 2019) and XLNet (Yang et al., 2019) have achieved excellent performance for some emotion classification tasks (Shen, Chen, Quan, & Xie, 2021; Zhou et al., 2020), BERT neglects dependency among predicted tokens and XLNet neglects positional information of the sentence.

To address the above problems, we provide a new solution for TEC by exploiting cascading broad learning (CBL) and sentence embedding using masked and permuted pre-training language model (MPNet) (Song, Tan, Qin, Lu, & Liu, 2020), named CBLMP. We adopt MPNet to generate sentence vectors for overcoming the shortcoming of BERT and XLNet and use the cascading of feature nodes (FNs) and enhancement nodes (ENs) to improve the performance of system by extending the general BL. In addition, We consider regularization parameter optimization for CBL. The main contributions of this work are listed as follows.

- We employ MPNet to generate sentence embeddings. It not only helps to capture the dependencies between words but also to obtain the positional information of words.
- We adopt cascading BL (CBL) to improve TEC performance with cascading FNs and cascading ENs.
- We use the L-curve method for regularization parameter optimization to maintain the balance between under-regularization and over-regularization to improve the CBL efficiency.

2. Related work

In the early works, they mainly aimed to explore bipartition (i.e., positive and negative) and tripartition (i.e., positive, negative, and neutral) of sentiment classification (Liu, 2015; Liu et al., 2020; Lv et al., 2021; Ran et al., 2018). Thus, sentiment classification can be regarded as bipartition-oriented or tripartition-oriented emotion classification. However, it neither considers the subtle emotional changes nor characterizes the users psychological state comprehensively (Peng et al., 2022).

So far, many researchers have started to use DL techniques to conduct TEC task. Based on CNN, Kim (2014) designed a neural architecture for TEC, and Zhang et al. (2018) adopted a multitask CNN for TEC. By using GRNN, Abdul-Mageed and Ungar (2017) designed an approach for the fine-grained emotion classification, and Tafreshi and Diab (2018) presented an architecture by using multitask learning to detect emotion.

By combining Bi-LSTM and attention, Basile et al. (2019) and Xiao (2019) proposed SymantoResearch and Figure Eight, respectively, for contextual emotion classification. Based on the BERT, Cao, Zhou, Yang, and Peng (2021) presented an approach for sentence-level TEC based on word-level emotion distribution; and Zhou et al. (2020)

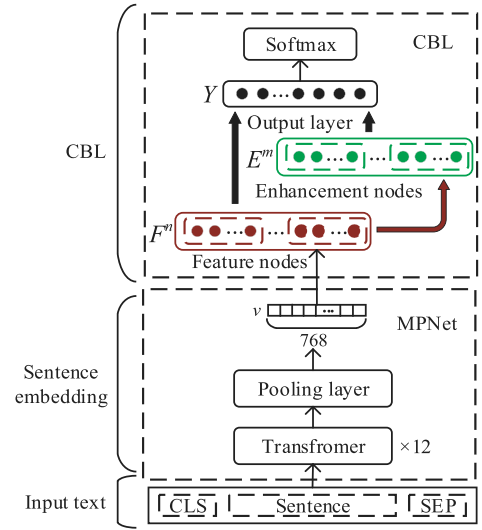


Fig. 1. The framework of CBLMP.

designed an emotional network (EmNet) for multi-label emotion classification through jointly learning emotion lexicons and carrying out classification.

Based on the adversarial learning, Li et al. (2023) proposed the EduEmo by using the elementary discourse units, BERT, and sparse attention; and Lin, Fu, Lin, and Wang (2022) proposed a method (i.e., MLEC-AM) for multi-label emotion classification with CNN and Bi-LSTM. In addition, on the basis of BL, Zeng et al. (2023) and Peng et al. (2023) explored the single-source cross domain and multi-source cross domain emotion classification, respectively.

Although there is strong learning capability for deep network models, it may consume large amounts of time and computing resources (Peng et al., 2022). As a flat network, BL has become an alternative solution for deep learning because of many advantages. So far, some researchers have employed BL for some applications like target location (Zhong & Liu, 2020), Fault diagnosis (Yu & Zhao, 2020), emotion recognition (Chen et al., 2022; Peng et al., 2021; Zhang, Wang, Xu, & Chen, 2019).

3. Modeling on TEC

In this section, we will provide a detailed description for the CBLMP. Firstly, the overall framework of our proposed CBLMP is provided and then each part of the model is described in detail, including MPNet for sentence embeddings, overview of original BL, cascading structure for BL and L-curve parameter optimization for selecting regularization parameter.

3.1. The overall framework of emotion classification model

The framework of CBLMP for TEC is shown in Fig. 1. In CBLMP, the texts are input into MPNet to obtain sentence embeddings, which is mapped into numerous FNs and ENs to obtain high dimensional linear features and non-linear features, respectively. Then, FNs and ENs are concatenated and input into the output layer to predict emotion.

3.2. Generating of sentence embeddings

MPNet is a pre-training language model improved from the masked language model (MLM) of BERT and the permutation language model (PLM) of XLNet. Firstly, MPNet consist of multiple Transformer encoder, and its multi head attention mechanism can help capture have long-term and short-term dependency; Secondly, the main idea of

MPNet is to combine the advantages of MLM and PLM in a unified framework. MLM can help the model obtain positional information of a sentence by utilizing masked word and position embedding. PLM can model dependencies between predicted words by using an autoregressive model.

The two-stream self-attention mechanism in PLM is utilized by MPNet for autoregressive prediction. This mechanism can capture dependency information between predicted words in each sentence. The idea of masking and positional encoding in MLM is also applied in pre-training by MPNet, which enables the model to obtain the position information of entire sentence when predicting masked words.

3.3. General broad learning

In general BL, sentence embeddings v are mapped into the FNs for extracting textual features and then FNs are transformed into the ENs to further extract the deep semantic features in a text. Then, both features extracted from FNs and ENs are input into the output layer to predict emotion.

Given a training set $\{X, Y\} \in \mathbb{R}^{N \times (M+C)}$, where N denotes the total number of texts, M denotes the dimension of sentence embeddings, and C denotes the total number of classes. Sentence embeddings X are linearly mapped into n groups of FNs. The i th group of FNs F_i is denoted by:

$$F_i = g(XW_{fi} + \beta_{fi}) \in \mathbb{R}^{N \times q}, i = 1, 2, \dots, n \quad (1)$$

where g denotes linear activation function, q denotes the number of FNs in each group, and β_{fi} and W_{fi} denote bias and weight matrices generated randomly, respectively.

Let $F^n \triangleq [F_1, F_2, \dots, F_n]$ denotes the matrix of n groups of FNs, which represents all linear features in a text. F^n are nonlinearly mapped into m groups of ENs. The j th group of ENs E_j is denoted by:

$$E_j = \varphi(F^n W_{ej} + \beta_{ej}) \in \mathbb{R}^{N \times r}, j = 1, 2, \dots, m \quad (2)$$

where φ denotes nonlinear activation function (e.g., sigmoid, tanh, ReLU), r denotes the total number of ENs in each group, and β_{ej} and W_{ej} are denote bias and weight matrices generated randomly, respectively.

Let $E^m \triangleq [E_1, E_2, \dots, E_m]$ denotes the matrix of m groups of ENs, which represents all the nonlinear features in a text. The matrix is concatenated by F^n and E^m for generating the actual input, which is denoted by:

$$A = [F^n, E^m] \in \mathbb{R}^{N \times (nq+mr)} \quad (3)$$

Thus, the final output of BL is denoted by:

$$\begin{aligned} \hat{Y} &= [g(XW_{f1} + \beta_{f1}), \dots, g(XW_{fn} + \beta_{fn}) \\ &\quad | \varphi(F^n W_{e1} + \beta_{e1}), \dots, \varphi(F^n W_{em} + \beta_{em})] W \\ &= [F_1, F_2, \dots, F_n | E_1, E_2, \dots, E_m] W \\ &= [F^n, E^m] W \\ &= AW \end{aligned} \quad (4)$$

The output weights W is denoted by:

$$W = A^+ Y \quad (5)$$

where A^+ denotes the pseudo inverse matrix of A .

However, calculating the pseudo inverse matrix directly will take a large amounts of time, and may lead to over-fitting. Ridge regression is a popular technique adopted for data modeling, which aims to solving the over-fitting problem often encountered in data analysis by introducing a regularization item. It can help to process the high-dimensional data and has good generalization ability. The main idea is to add a perturbation (i.e., a penalty item), on the basis of the original least square estimation, which makes the generalized inverse is easy to solve when it fail to obtain a solution.

Thus, the ridge regression is utilized as an objective function to solve W , which can be denoted by:

$$\argmin_W (\|Y - AW\|_2^2 + \lambda \|W\|_2^2) \quad (6)$$

where λ denotes regularization parameters.

On the basis of the regularized least square theory, W is denoted by:

$$W = (\lambda I + AA^T)^{-1} A^T Y \quad (7)$$

Notably, weights of middle layers W_{fi} , W_{ej} , β_{fi} , and β_{ej} are fixed, and only the output weights W need to be updated, which is different from the end-to-end learning manner of deep learning. The BL learning process is provided in Algorithm 1.

Algorithm 1 The BL learning process

Input: Sentence embeddings X

Output: Output weights W

```

1: for  $i = 1; i \leq n$  do
2:   Random  $W_{fi}, \beta_{fi}$ 
3:   Calculate  $F_i = g(XW_{fi} + \beta_{fi})$ 
4: end for
5: Set the FNs  $F^n \triangleq [F_1, F_2, \dots, F_n]$ 
6: for  $j = 1; j \leq m$  do
7:   Random  $W_{ej}, \beta_{ej}$ 
8:   Calculate  $E_j = \varphi(F^n W_{ej} + \beta_{ej})$ 
9: end for
10: Set the ENs  $E^m \triangleq [E_1, E_2, \dots, E_m]$ 
11: Set  $A = [F^n, E^m]$  and calculate output weights  $W$ 

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On the basis of the analysis on Algorithm 1, the time complexity of BL is $O(Mnq + Mmr)$.

3.4. Cascading broad learning

General BL is a feed-forward network of single-layer mapping. It only generates FNs and ENs through one layer of linear mapping and nonlinear mapping, which makes it difficult to capture high-level abstract features for TEC. To solve the above problem, we adopt CBL for performing multi-layer mapping of features to improve the BL ability.

In CBL, sentence embeddings are linearly mapped into multiple FN groups in cascading order, and then all FNs are nonlinearly mapped into multiple EN groups in cascading order. Finally, both features extracted from FNs and ENs are input into the output layer to obtain the emotional probability distribution. The solution of output weight is the same as BL. The specific structure of CBL is shown in Fig. 2.

Given a training set $\{X, Y\} \in \mathbb{R}^{N \times (M+C)}$, sentence embedding X is linearly mapped into the first group of FNs F_1 , which is denoted by:

$$F_1 = g(XW_{f1} + \beta_{f1}) \triangleq g^1(X; \{W_{f1}, \beta_{f1}\}) \in \mathbb{R}^{N \times q} \quad (8)$$

Then, FNs F_1 are linearly mapped into the second group of FNs F_2 , which is denoted by:

$$\begin{aligned} F_2 &= g(F_1 W_{f2} + \beta_{f2}) \\ &= g(g(XW_{f1} + \beta_{f1}) W_{f2} + \beta_{f2}) \\ &\triangleq g^2(X; \{W_{fi}, \beta_{fi}\}_{i=1,2}) \in \mathbb{R}^{N \times q} \end{aligned} \quad (9)$$

Repeating the above process to generate FNs, the k th group of FNs F_k is denoted by:

$$\begin{aligned} F_k &= g(F_{k-1} W_{fk} + \beta_{fk}) \\ &\triangleq g^k(X; \{W_{fi}, \beta_{fi}\}_{i=1}^k) \in \mathbb{R}^{N \times q}, k = 2, 3, \dots, n \end{aligned} \quad (10)$$

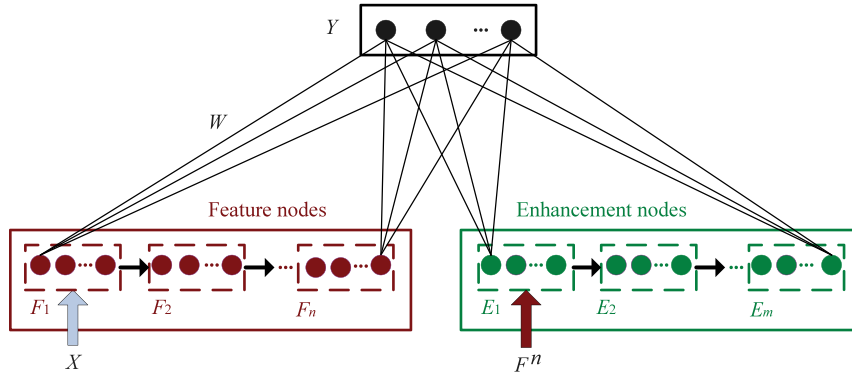


Fig. 2. The structure of CBL.

Thus, $F^n \triangleq [F_1, F_2, \dots, F_n]$ is concatenated by n groups of FNs. It is utilized to nonlinearly map into the first group of ENs E_1 , which is denoted by:

$$E_1 = \varphi(F^n W_{e1} + \beta_{e1}) \triangleq \varphi^1(F^n; \{W_{e1}, \beta_{e1}\}) \in \mathbb{R}^{N \times r} \quad (11)$$

In addition, E_1 is utilized to nonlinearly map into the second group of ENs E_2 , which is denoted by:

$$\begin{aligned} E_2 &= \varphi(E_1 W_{e2} + \beta_{e2}) \\ &= \varphi(\varphi(F^n W_{e1} + \beta_{e1}) W_{e2} + \beta_{e2}) \\ &\triangleq \varphi^2(F^n; \{W_{ej}, \beta_{ej}\}_{j=1,2}) \in \mathbb{R}^{N \times r} \end{aligned} \quad (12)$$

Repeating the above process to generate ENs, the u th group of ENs E_u is denoted by:

$$\begin{aligned} E_u &= \varphi(E_{u-1} W_{eu} + \beta_{eu}) \\ &\triangleq \varphi^u(F^n; \{W_{ej}, \beta_{ej}\}_{j=1}^u) \in \mathbb{R}^{N \times r}, u = 2, 3, \dots, m \end{aligned} \quad (13)$$

Thus, $E^m \triangleq [E_1, E_2, \dots, E_m]$ is concatenated by m groups of ENs. Then, F^n and E^m are concatenated to construct $A = [F^n, E^m]$ as the actual input. Thus, the output of CBLMP is denoted by:

$$\begin{aligned} \hat{Y} &= [g^1(X; \{W_{f1}, \beta_{f1}\}), \dots, g^n(X; \{W_{fi}, \beta_{fi}\}_{i=1}^n) \\ &\quad | \varphi^1(F^n; \{W_{e1}, \beta_{e1}\}), \dots, \varphi^m(F^n; \{W_{ej}, \beta_{ej}\}_{j=1}^m)] W \\ &= [F_1, F_2, \dots, F_n | E_1, E_2, \dots, E_m] W \\ &= [F^n, E^m] W \\ &= A W \end{aligned} \quad (14)$$

where W is calculated by Eq. (7).

The CBLMP learning process is listed in Algorithm 2.

According to Algorithm 2, the time complexity of CBLMP is $O(Mnq + Mmr)$. The time complexity of SVM, TextCNN, RCNN, and Bi-LSTM is $O(M^2)$, $O(uvlM^2)$, $O(uvlM^2)$, and $O(lM^2)$, respectively, where l denotes the sequence length, u denotes the number of convolution layers, and v denotes the kernel size of convolution.

In BL, m , n , q , and r are set to a small value, respectively. In general, they are smaller than l , and M is far larger than m , n , q , and r . Thus, the time complexity of CBLMP is smaller than that of SVM, TextCNN, RCNN, and Bi-LSTM.

3.5. Selection of regularization parameter using L-curve

Since regularization can prevent over-fitting effectively and can improve the generalization ability of model, it has been widely exploited in ML. As shown in Eq. (6), the objective function of BL is ridge regression, which prevent falling into a global optimal solution of least squares by adding penalty term $\lambda \|W\|_2^2$. However, the estimation results of ridge regression are quite different under different regularization parameters. If λ is set too small, it will be more prone to over-fitting

Algorithm 2 CBLMP learning process

Input: Sentence embeddings X

Output: Output weights W

- 1: Random W_{f1}, β_{f1}
- 2: Calculate $F_1 = g(XW_{f1} + \beta_{f1})$
- 3: **for** $k = 2; k \leq n$ **do**
- 4: Random W_{fk}, β_{fk}
- 5: Calculate $F_k = g(F_{k-1}W_{fk} + \beta_{fk})$
- 6: **end for**
- 7: Set the FNs $F^n \triangleq [F_1, F_2, \dots, F_n]$
- 8: Random W_{e1}, β_{e1}
- 9: Calculate $E_1 = \varphi(F^n W_{e1} + \beta_{e1})$
- 10: **for** $u = 2; j \leq m$ **do**
- 11: Random W_{eu}, β_{eu}
- 12: Calculate $E_u = \varphi(E_{u-1}W_{eu} + \beta_{eu})$
- 13: **end for**
- 14: Set the ENs $E^m \triangleq [E_1, E_2, \dots, E_m]$
- 15: Set $A = [F^n, E^m]$ and calculate output weights W

because of large variance and small bias. If is set too large, the error of the estimation will be large because of large bias and small variance. Thus, it is essential to select a suitable λ to ensure the classification performance of CBLMP.

The L-curve (Hansen, 1992) method is an effective and suitable method for regularization parameter optimization. First, the norm of residuals $\|Y - AW\|_2^2$ and the norm of regularized solution $\|W\|_2^2$ are calculated under different λ . Then, all of data points $(\log(\|Y - AW\|_2^2), \log(\|W\|_2^2))$ are fitted with a smooth decreasing curve, which has an obvious inflection point connecting the part of large slope and the part of small slope. Its overall shape is similar to "L". Finally, the value corresponding to inflection point is selected as the final result for λ . Specifically, at the stage of large slope, the slight increase in $\|Y - AW\|_2^2$ will make $\|W\|_2^2$ drop significantly, which will cause the model to fall into the stage of under-regularization. In contrast, an increase in $\|W\|_2^2$ will make $\|Y - AW\|_2^2$ drop significantly at the stage of a small slope, which will cause the model to fall into the stage of over-regularization. The parameter at the inflection point can better maintain a balance between them.

In the experiment, it is assumed that the distribution of the validation set and test set are equal. L-curve is adopted to determine the best regularization parameter and then evaluate it on the test set.

4. Performance evaluation

In this section, we will provide a detailed introduction of the datasets, baseline methods, parameter settings, and evaluation metrics utilized in our experiment, and provide the comparison analysis among

Table 1
Statistics of the datasets.

Dataset	Emotion	Train	Validation	Test
SemEval	Happy	5191	180	369
	Sad	6357	151	308
	Angry	6027	182	324
	other	12 585	2242	4508
Usual	Happy	5379	391	1018
	Sad	4990	346	900
	Angry	8344	586	1508
	Fear	1220	87	210
	Surprise	2086	170	373
	Neutral	5749	420	990
Virus	Happy	4423	923	1540
	Sad	649	165	219
	Angry	1322	314	463
	Fear	555	75	190
	Surprise	197	17	68
	Neutral	1460	476	520

CBLMP with the baseline methods. In addition, we carry out experiments to further verify the CBLMP efficiency and to analyze the impact on our proposed method from different pre-trained language model. The experimental devices include a laptop with Intel-i5-9400 2.9 GHz CPU, 16 GB memory and Nvidia 1660Ti GPU.

4.1. Datasets

To verify the effectiveness of CBLMP, we carry out comparative experiments on the English text dataset from SemEval-2019 Task 3: EmoContext contextual emotion detection in the text, which is provided by Chatterjee, Narahari, Joshi, and Agrawal (2019), and the Chinese text dataset from the evaluation of Weibo emotion classification technology in the ninth China national conference on social media processing (SMP2020-EWECT)¹, which includes two kinds of data: usual and virus. Detailed information for each kind of dataset is shown in Table 1.

Since the SemEval dataset is extracted from online text, it contains emoji, informal words and irregular grammar, etc., which are adverse to the generating of sentence embedding. Thus, Ekphrasis toolkit (Rohanian, Taslimipoor, Evans, & Mitkov, 2018) is utilized to process the text, the specific process including: (i) Emoticons are changed into the relevant emotional words (e.g., “:-” is changed into “Unhappyface”); (ii) Informal words are transformed into the relevant correct words (e.g., “Thks” is changed into “Thanks”); and (iii) Capital letters are changed into lowercase letters. In addition, Ekphrasis toolkit is also employed to process the Usual and Virus dataset because of much invalid information: (i) Remove url, email and “@” token; (ii) Traditional Chinese character and capital letters are changed to simplified Chinese character and lowercase letters respectively; and (iii) Emoji are changed into the corresponding emotional word.

4.2. Evaluation metrics

We adopt the official evaluation metrics (i.e., accuracy and Macro-F1). Accuracy refers to the ratio of the correct classified samples to the total number of samples, which can be denoted by:

$$Accuracy = \frac{\sum_{c=1}^C (TP_c + TN_c)}{\sum_{c=1}^C (TP_c + FP_c + TN_c + FN_c)} \quad (15)$$

where TP_c , FP_c , TN_c , and FN_c denote the number of true positive, false positive, true negative, and false negative of the c th emotion class, respectively.

Macro-F1 refers to the average of F1 values of each class. The F1 value is the weighted average of precision and recall. Macro-F1 can objectively reflect the performance of the model under class imbalance. The specific is listed as follows:

$$F1_c = \frac{2 \times P_c \times R_c}{P_c + R_c} \quad (16)$$

$$Macro - F1 = \frac{1}{C} \sum_{i=1}^C F1_c \quad (17)$$

where P_c , R_c and $F1_c$ denote precision, recall, and F1 values of the c th emotion class, respectively.

4.3. Baseline methods

To verify the effectiveness of CBLMP in TEC, we compare it with the following baseline models:

- **SVM** (Cao et al., 2020): It is a commonly adopted as a baseline model for TEC task, which classifies two classes by optimal decision boundary. In the experiment, “one-versus-one” is applied to train a classifier for every two classes.
- **TextCNN** (Kim, 2014): It is a textual classification model that utilizes multi-convolutional kernels with different sizes to combine automatically the n -gram features, so as to capture deep-level semantic information.
- **RCNN** (Girshick, Donahue, Darrell, & Malik, 2014): It is a text classification model which adopts a bidirectional recurrent neural network to extract contextual features and uses a maximum pooling layer to capture key features.
- **Bi-LSTM** (Schuster & Paliwal, 1997): It treats the sentence as a sequence and input into Bi-LSTM to extract contextual features. It can capture the long-distance dependencies of text.
- **Bi-LSTM-ATTN** (Cao et al., 2020): It adds an attention layer in the Bi-LSTM. The attention mechanism will help to capture the local features of a text.
- **ERNIE-CNN**²: It is an emotion classification model developed for SMP2020-EWECT. The pre-trained language model ERNIE is utilized as the embedding layer and a convolutional neural network is utilized for feature extraction.
- **ERNIE-RNN**: It similar to ERNIE-CNN, the difference is that recurrent neural network is utilized to extract contextual feature.
- **SNU-IDS** (Bae, Choi, & goo Lee, 2019): It is ranked 7th method of SemEval-2019 Task 3 proposed by Bae et al. (2019). It achieves a Micro-F1 score of about 76.60% on the final evaluation.
- **YUN-HPCC** (Li, Wang, & Zhang, 2019): It is ranked 10th method of SemEval-2019 Task 3 proposed by Li et al. (2019). It achieves a Micro-F1 score of about 75.88% on the final evaluation.
- **SCIA** (Rebiai, Andersen, Debrenne, & Lafargue, 2019): It improves performance by the ensemble with four different models: contextual Bi-LSTM, categorical Bi-LSTM, binary convolutional Bi-LSTM, and GRU.
- **DLCF-DCA** (Xu et al., 2022): It consists of DLCF and DCA, where DLCF aims to capture local context based on max distance and DCA aims to pay attention to the cluster which is important for classification.
- **ASC-pos-wt** (Ma, Zhang, & Song, 2021): It is a mechanism to capture positional bias, namely positional-biased weight, which can be utilized to enhance the classification performance in BERT.
- **XLNet** (Yang et al., 2019): It is a pre-training language model using the pre-trained strategy of permutation language model.

¹ <https://smp2020ewect.github.io/>

² <https://github.com/smp2020ewect/smp2020ewect.github.io/tree/master/reports>

Table 2

Parameters for training deep networks.

Model	Parameter	All dataset
TextCNN	Kernel size	3/4/5
	Activation function	ReLU
RCNN	Bi-LSTM layer	1
	Bi-LSTM unit	400
	Kernel size	2
	Activation function	ReLU
Bi-LSTM	Bi-LSTM layer	2
	Bi-LSTM unit	400
Bi-LSTM-ATTN	Bi-LSTM layer	1
	Bi-LSTM unit	400
	Attention layer	1

- **ELECTRA** (Clark, Luong, Le, & Manning, 2020): It is a pre-training language model that utilized replaced token detection pre-trained strategy and consists of a generator and a discriminator.
- **MLEC-AM** (Lin et al., 2022): It is a method that adopted the adversarial multi-task learning for textual emotion classification.

4.4. Parameter setting

For a fair comparison, these DL methods (e.g., TextCNN, RCNN, Bi-LSTM, Bi-LSTM-ATTN) utilize word embedding generated by MPNet pre-trained on Chinese text with dimension of 768. In SemEval dataset, three rounds of dialogue are connected by the token “[SEP]” and input into MPNet pre-trained on English text. We set sufficient epochs and introduce the early stopping (Caruana, Lawrence, & Giles, 2001) to ensure that the model can converge to a suitable position. A proper learning rate is utilized to avoid model falling into a local optimum at an early stage. Thus, an optimization strategy of the learning rate is utilized to obtain a well-performed model. These DL-based methods are implemented in the tensorflow framework and their hyperparameter settings are shown in Table 2.

In addition, BL, CBLMP and SVM all take sentence embedding generated by MPNet with dimension of 768. The sentence embedding is the output of the pooling layer in MPNet. As to SemEval, the three sentence embeddings corresponding to three dialogues are concatenated as the input of the model. We adjust the hyperparameters of proposed model according to the actual performance of the validation set, and then determine the best set of parameters for testing. The hyperparameters that need to be adjusted mainly include the number of FNs and ENs, and regularization parameters.

To explore the influence from the number of FNs and ENs, the number of FNs per group k and the number of ENs per group q of CBLMP are set to 50, the number of FN groups n and the number EN groups m are set to [1, 4, 7, ..., 28]. The Macro-F1 score with different n and m is shown in Fig. 3.

From Fig. 3, it can be seen that Macro-F1 increases significantly when n increases from 1 to 28. Given a certain n , Macro-F1 increase slightly when m increases. This shows that FNs help to extract main information of text, and the ENs further improve performance. Thus, we can determine the number of groups corresponding to the best Macro-F1 as the final result, that is, $n = 22$ and $m = 19$.

We use the L-curve to determine the proper regularization parameter λ , set λ [0.000005, 0.00001, ..., 0.1] for CBLMP, and provide the corresponding L-curve on the validation set. The L-curve is shown in Fig. 4.

From Fig. 4, it is found that the value near the inflection point is between 0.005 and 0.0005, and the model is at the equilibrium between under regularization and over regularization. When λ is greater than 0.005, the model is at over regularization state. When λ is less than 0.0005, the model is at under regularization state. Thus, we select the parameter λ near the inflection point, i.e., $\lambda = 0.0005$. Macro-F1 under different λ on test set is shown in Fig. 5.

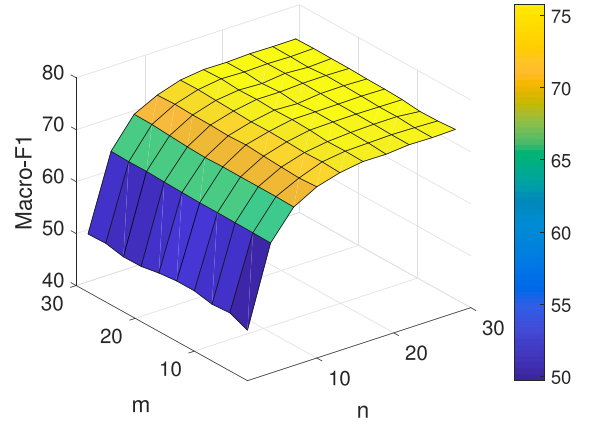


Fig. 3. Macro-F1 under different number of FNs and ENs for CBLMP.

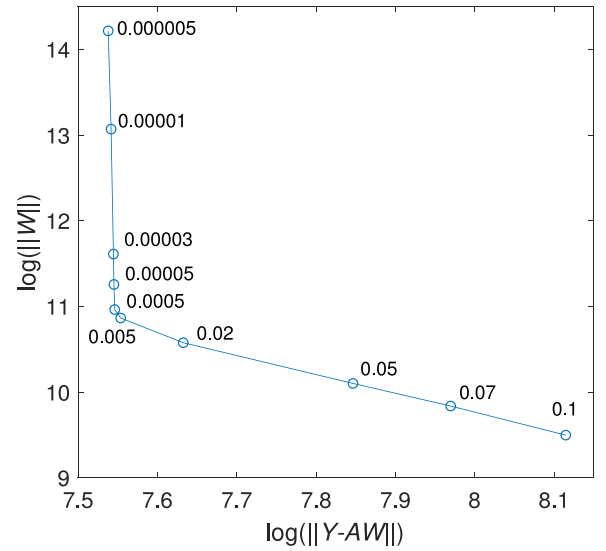


Fig. 4. The L-curve of CBLMP on Usual dataset.

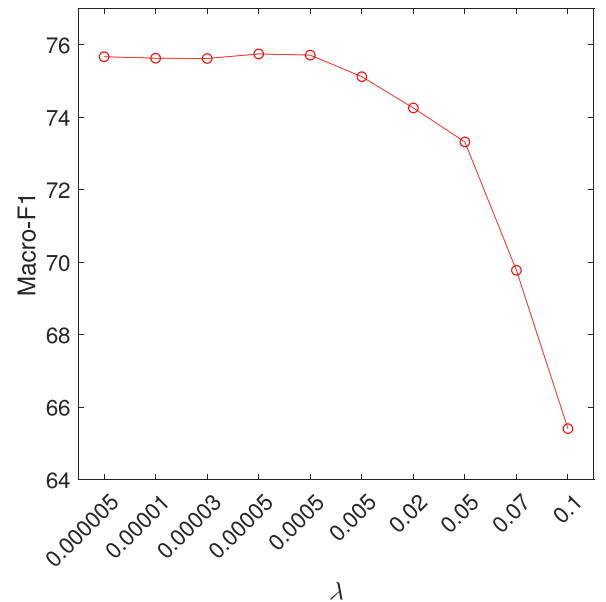
Fig. 5. Macro-F1 of different λ for CBLMP on Usual dataset.

Table 3

Related parameters for BLMP and CBLMP.

Model	Parameter	SemEval	Usual	Virus
BLMP	(n, k)	(41, 50)	(22, 50)	(20, 50)
	(m, q)	(41, 50)	(19, 50)	(15, 50)
	λ	0.0005	0.0003	0.0003
	Activation function	tanh	tanh	tanh
CBLMP	(n, k)	(41, 50)	(22, 50)	(18, 50)
	(m, q)	(45, 50)	(19, 50)	(15, 50)
	λ	0.0005	0.0005	0.0001
	Activation function	tanh	tanh	tanh

Table 4

Performance of each method on Usual.

Model	Acc	Macro-F1
SVM	72.43	66.13
TextCNN	76.16	72.85
RCNN	74.90	70.86
Bi-LSTM	76.24	73.11
Bi-LSTM-ATTN	76.83	73.94
ERNIE-CNN	76.77	72.90
ERNIE-RNN	77.37	73.42
DLCF-DCA	76.41	73.37
Pos-wt	75.85	72.33
XLNet	76.96	73.63
ELECTRA	77.20	74.61
MLEC-AM	77.17	74.48
BLMP	77.79	75.25
CBLMP	78.54	75.79

Table 5

Performance of each method on Virus.

Model	Acc	Macro-F1
SVM	70.42	55.49
TextCNN	77.53	63.29
RCNN	76.03	57.57
Bi-LSTM	77.40	60.66
Bi-LSTM-ATTN	78.06	61.95
ERNIE-CNN	77.65	64.56
ERNIE-RNN	77.15	63.60
DLCF-DCA	78.66	64.20
Pos-wt	75.08	59.54
XLNet	77.75	64.55
ELECTRA	78.66	64.83
MLEC-AM	78.14	64.35
BLMP	77.88	64.41
CBLMP	79.03	65.19

From Fig. 5, it can be seen that Macro-F1 increases slightly when λ increases from 0.000005 to 0.00005 and reach peak when $\lambda = 0.00005$. When λ increases from 0.0005 to 0.1, Macro-F1 decreases significantly. It shows that over-regularization will not fit the train data properly and the estimation error will be large.

Thus, the hyperparameters of BL+MPNet (BLMP) and CBLMP on each dataset are show in Table 3.

4.5. Comparison with baseline methods

The performance of the baseline methods and CBLMP on three kinds of data sets are shown in Table 4, Tables 5 and 6, respectively.

From Tables 4–6, it is found that CBLMP outperforms the baseline methods on three different datasets. On Usual dataset, the accuracy and Macro-F1 of CBLMP outperform ERNIE-RNN by 1.16% and 2.37%, respectively. On Virus dataset, the accuracy and Macro-F1 of CBLMP outperform DLCF-DCA by 0.37% and 0.99%, respectively. On SemEval dataset, the accuracy and Macro-F1 of CBLMP outperform YUN-HPCC by 0.83% and 0.62%.

We also observe that CBLMP can improve the TEC performance more effectively than BLMP on three kinds of datasets. This indicates

Table 6

Performance of each method on SemEval.

Model	Acc	Macro-F1
SVM	82.52	57.39
TextCNN	89.29	73.51
RCNN	88.76	68.03
Bi-LSTM	87.12	68.34
Bi-LSTM-ATTN	87.70	69.45
SNU-IDS	92.85	77.13
YUN-HPCC	89.31	75.16
SCIA	89.71	75.66
DLCF-DCA	88.07	68.22
Pos-wt	86.98	65.27
XLNet	88.20	74.29
ELECTRA	88.92	75.08
MLEC-AM	87.70	73.69
BLMP	89.53	74.80
CBLMP	90.14	75.78

Table 7

Training time (s) of each method on three datasets.

Model	Usual	Virus	SemEval
TextCNN	2012.27	946.99	2370.25
RCNN	2440.22	1031.54	2547.31
Bi-LSTM	4481.99	978.03	5013.67
Bi-LSTM-ATTN	6873.24	1728.66	6758.3
BLMP	199.58	65.48	226.44
CBLMP	193.23	63.3	219.73

Table 8

Inference time (s) of each method on three datasets.

Model	Usual	Virus	SemEval
TextCNN	279.31	94.50	305.33
RCNN	292.84	103.29	321.18
Bi-LSTM	327.02	115.42	338.73
Bi-LSTM-ATTN	314.89	119.46	323.25
DLCF-DCA	369.77	135.51	377.63
Pos-wt	250.39	82.04	283.42
BLMP	192.63	63.38	217.54
CBLMP	188.74	61.02	210.13

that the multi-layer mapping of cascading structure can help the model extract more useful information for TEC. However, SNU-IDS outperforms CBLMP with 2.71% and 1.35% in Acc and Macro-F1 on SemEval, respectively. The main reason is that SNU-IDS adopts two stacked Bi-LSTM encoders and two pre-training language models (i.e., word2vec and ELMo).

In addition, we compare the training time and inference time on the test set among baseline methods and CBLMP. The training time includes the time of generating word embeddings and training the downstream model, and The inference time include the time of generating word embedding and inference process. The species are listed in Tables 7 and 8, respectively.

From Table 7, it can be observed that the training time of CBLMP is less than that of the baseline methods. The main reason is that the DL-based methods require multiple rounds of training to optimize network parameters, while CBL only needs to optimize the weights of the output layer through the ridge regression. From Table 8, it can be seen that the inference time of CBLMP is less than that of the baseline methods. In addition, the training time of CBLMP is shorter than that of BLMP. The above results indicate that CBLMP can not only improve performance but also reduce computational complexity.

4.6. Effect of FNs and ENs

To analyze the effect FNs and ENs on CBLMP, we first set the number of nodes in each group to 50, and conduct the performance comparison experiment on the Usual dataset, according to the following

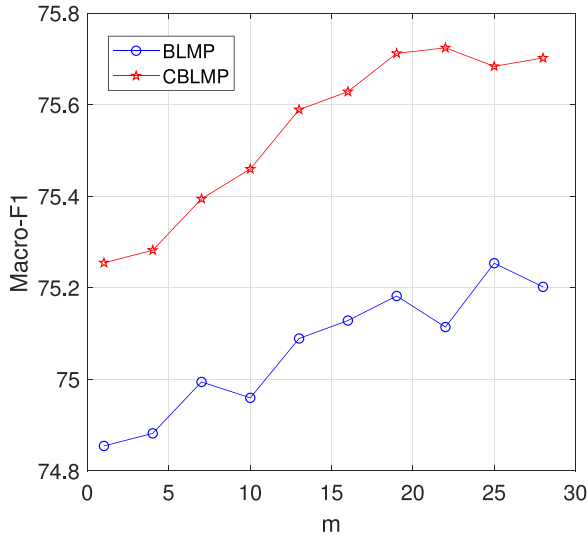


Fig. 6. The effect under different number of groups of ENs on Usual dataset.

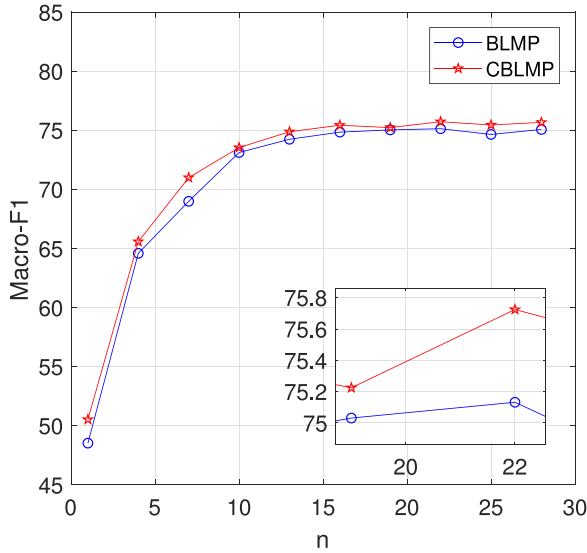


Fig. 7. The effect under different number of groups of FNs on Usual dataset.

two cases: (i) when $n = 22$, m changes from 1 to 28, and the effect on CBLMP is analyzed under the different number of groups of ENs; and (ii) when $m = 19$, n changes from 1 to 28, and the effect on CBLMP is analyzed under the different number of groups of FNs. The experimental results are shown in Fig. 6 and Fig. 7, respectively.

From Fig. 6, it can be seen that the Macro-F1 score increases slightly as m increases, which illustrates that the performance is improved but very limited by adding the number of groups for the ENs. From Fig. 7, it can be seen that the Macro-F1 score of CBLMP significantly increases with the increase of n from 1 to 13, which shows the cascading FNs are more effective than the cascading ENs. One reason may be that a small number of FNs is difficult to capture sufficient useful information. The other reason is that FNs are responsible for capturing features from texts, while ENs are responsible for further improving the CBL performance. In addition, it can be observed that when $n = 22$ and $m = 19$, CBLMP can achieve the best performance.

4.7. Effect of pre-trained language model

The embedding representation for text is an important factor for the TEC performance. A suitable pre-trained language model can better

Table 9

Performance of each pre-trained language model on Usual dataset.

Pre-trained language model	Acc	Macro-F1
Word2Vec	63.49	53.41
GloVe	71.34	65.79
ELMo	72.76	68.89
BERT	77.52	75.06
MPNet	78.54	75.79

Table 10

Performance of each pre-trained language model on Virus dataset.

Pre-trained language model	Acc	Macro-F1
Word2Vec	67.91	48.34
GloVe	72.67	53.16
ELMo	73.47	54.20
BERT	78.10	64.53
MPNet	79.03	65.19

Table 11

Performance of each pre-trained language model on SemEval dataset.

Pre-trained language model	Micro-F1
Word2Vec	59.25
GloVe	63.03
ELMo	66.72
BERT	73.31
MPNet	74.67

obtain sentence vectors. To verify the effect on the CBLMP performance under different pre-trained language models are adopted as the sentence embeddings of CBLMP.

- **Word2Vec** (Mikolov, Chen, Corrado, & Dean, 2013): It maps sparse one-hot word embedding to dense word embedding by using three-layer network. This method includes CBOW and Skip-Gram. In CBOW, the context is utilized to predict the current word, and in Skip-Gram, the current word is utilized to predict context. In the experiment, all of the word embeddings with a dimension of 200 are averaged as sentence embedding.
- **GloVe** (Pennington, Socher, & Manning, 2014): It is a word representation method on the basis of word frequency statistics, which uses global and local statistical information of words to generate word embedding. In the experiment, all of the word embeddings with dimension of 300 are averaged as sentence embedding.
- **ELMo** (Peters et al., 2018): It generates dynamic word vectors by using multi-layer Bi-LSTM. In the experiment, all of the word embeddings with dimensions of 1024 are averaged as sentence embedding.

The accuracy score and Macro-F1 of each pre-trained language model are shown in Table 9, Tables 10 and 11, respectively.

From Table 9, Tables 10 and 11, it can be seen that MPNet can achieve the best results for these three datasets. The method using MPNet to generate word embedding is outperforms BERT. This illustrates that MPNet can help the downstream model to classify emotion in text better than BERT. Word2Vec and GloVe perform the worst among of them, because they are unable to solve the sensitivity of word order in emotion classification, for example, the static sentence embedding of “I like you” vs “I don’t hate you” and “I hate you” vs “I don’t like you” are the same, but different word orders make them express different emotions.

4.8. Discussion

In general, CBLMP performs better than most baseline methods, due to the advantages of CBL structure and MPNet. These advantages

mainly include: (i) Cascading FNs and ENs can help to capture textual features and enhance model performance, respectively; (ii) MPNet can further help CBL to capture semantic features and contextual features; (iii) Solving pseudo-inverse matrix through the ridge regression can not only prevent over-fitting but also reduce the consumption of computing source and training time.

However, comparing the SOTA models proposed by SMP2020-EWECT and SemEval 2019, respectively, we find that these models outperform CBLMP by assembling multiple techniques and using a more complex framework. For example, BERT Forest is ranked 1st and proposed by Li et al. from Tencent Oteam et al. in SMP2020-EWECT,³ which can achieve Acc = 80.85% and Macro-F1 = 78.56% on Usual, and Acc = 81.10% and Macro-F1 = 69.76% on Virus; NELEC and ANA are ranked 1st and 5th and proposed by Agrawal and Suri (Agrawal & Suri, 2019) and Huang et al. (Huang, Trabelsi, & Zaiane, 2019), respectively, in SemEval 2019. Since Macro-F1 is adopted in CBLMP, we conduct an experiment for NELEC and ANA, respectively, via ACC and Macro-F1, it can achieve Acc = 93.72% and Macro-F1 = 77.95%, and Acc = 93.04% and Macro-F1 = 77.47%, respectively. According to the above comparison, it can be seen that BERT Forest outperforms CBLMP 2.34% and 1.77% in Acc and Macro-F1 on Usual, respectively, and outperforms CBLMP 2.07% and 4.57% in Acc and Macro-F1 on Virus, respectively. In addition, NELEC outperforms CBLMP with 3.58% and 2.17% in Acc and Macro-F1 on SemEval, respectively, and ANA outperforms CBLMP with 2.9% and 1.69% in Acc and Macro-F1 on SemEval, respectively.

The main reason is that BERT Forest utilizes a large number of Chinese pre-trained models for pre-training, adopts different training strategies for fine-tuning, to obtain a large number of basic features with low deviation and high square error, and uses the stacking scheme on these features; NELEC utilizes lexical features and word embeddings from pre-trained neural models to boost performance; ANA encodes the utterance via GloVe, ELMo, and DeepMoji embeddings, following which a contextual LSTM is utilized to encode the entire dialogue for prediction.

However, since CBLMP only adopts one pre-trained model MPNet, rather than adopting more pre-trained models to generate word embeddings. It is obvious that the framework of CBLMP is more simple than those of SNU-IDS, ANA and BERT Forest. Due to the simple structure of CBLMP, after reaching a certain level, it is difficult to improve its performance further.

5. Summary and future work

In this paper, we have provided an overview for the current research status in TEC and proposed a method CBLMP for TEC to address the problems (e.g., training time is long, and theoretical analysis of DL models is difficult), by integrating MPNet and CBL. In CBLMP, we have adopted MPNet to generate sentence embedding and utilized CBL to capture textual features and predict emotion. At last, we have carried out many experiments on the public datasets by comparing it with the baseline methods. It is found that CBLMP can better improve TEC performance according to the experimental results.

Although BL outperforms simple machine learning models in TEC tasks, large-scale complex deep learning models still outperform BL. The reason is that BL still finds it difficult to learn deeper semantic information in text data. As to our future work, we will provide an optimization scheme for BL to improve the TEC performance and further explore to adopt other datasets for verifying the effectiveness of CBLMP. We also plan to explore the cross-domain and multimodal TEC tasks by using CBLMP.

CRedit authorship contribution statement

Lihong Cao: Writing – review & editing. **Rong Zeng:** Methodology, Software. **Sancheng Peng:** Funding acquisition, Supervision, Writing – original draft. **Aimin Yang:** Funding acquisition. **Jianwei Niu:** Supervision. **Shui Yu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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³ <https://pan.baidu.com/s/1iSxAKETd2Ia2f3uvsTQLrA#list/path=%2F>

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